KALID IBRAHIM

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Kalid I.

OBJECTIVE

UMBC CE sophomore seeking an Embedded/Software internship to build reliable, efficient systems and learn from experienced teams. Strong interest in EE (digital design , PCB/FPGA, and hardware–software integration).

EDUCATION

- **University of Maryland, Baltimore County — B.S. Computer Engineering** *Expected May 2028*
  GPA: 3.8/4.0 — Relevant: CMSC 201/202, CMPE 212 (Digital Design)

TECHNICAL SKILLS

- **Languages:** Python, C++, MATLAB(basic)
- **Embedded / Digital:** Arduino; GPIO, PWM, analog sensors; digital logic design (K-maps, combinational and sequential circuits, counters/FSMs); Verilog(Structural Design and Testbench)
- **Tools:** Linux, Git/GitHub, Docker (basics), Oscilloscope, Breadboard Circuits
- **Data / ML:** pandas, NumPy, basic PyTorch

SOFT SKILLS

- **Communication & Teamwork:** Clear handoffs, active listening, and cross-role coordination (camp counselor, pantry).
- **Customer Service:** Patient, solution-oriented support under time pressure (Hallmark).
- **Problem Solving & Debugging:** Break issues into steps; try different scenarios to see where things might fail; document assumptions.
- **Accountability & Time Management:** Take ownership of outcomes, show up prepared, and balance classes, work, clubs and projects.
- **Leadership & Conflict Resolution:** Lead small groups, de-escalate, and keep activities on schedule (camp).
- **Adaptability:** Learn new tools quickly (from scripts to structured OOP; basic CI; Git/GitHub workflows).

PROJECTS

- **NLP Book Recommender ("ReadMe") - Python/Flask/React** *2025*
  - Combined content features with k-NN and TF-IDF/Sentence-BERT embeddings; cosine similarity.
  - Built a simple React UI; evaluated ranking with Precision@K; experiments versioned in Git.
- **H2-ROVER Autonomous Vehicle - Arduino/C++** *2025*
  - Built a hydrogen fuel cell–powered line-following rover with motor driver and environmental sensors. - Tuned design to achieve the highest cost-effectiveness and top raw score in my lab section.
- **3-Bit to 7-Segment Display Decoder – Verilog/Digital Logic** *2025*
  - Designed a 3-bit binary to 7-segment display decoder: built truth table, simplified with K-maps, implemented structural Verilog with basic gates, and verified digit outputs in simulation and breadboard hardware.
- **Song Player - C++** *2025*
  - Program that loads songs from a file into a templated linked-list playlist that supports search, add, and sort by Spotify streams with safe dynamic memory management.
- **RetrieverBell Network - Python** *2025*
  - Switch-network graph with add/remove nodes, trunk connections, call start/end; BFS/DFS; JSON save/load.

EXPERIENCE

- **Pantry Worker - Retriever Essentials (UMBC)** *Aug 2025 – Present*
  Support campus food access: inventory, customer service, and data tracking in a fast-paced student-facing environment.
- **Sales Associate - Hallmark** *Apr 2023 – May 2024*
  Assisted customers, handled register, and helped train new staff; managed restocking and merchandise displays.
- **Camp Counselor - Coffield Rec Center** *Jun 2024 – Aug 2025*
  Led daily activities for 16 campers; ensured safety and resolved conflicts; collaborated with staff on schedules.

AWARDS & ACTIVITIES

- **Clubs:** ***EESA*** – Cabinet Officer | ***NSBE*** – Events | ***MIF*** – Public Relations | ***IEEE*** – Student Member
- **Interests**: embedded systems, computer architecture, ML systems.